



KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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SDC1

II-I CSE, CSE(AI&ML) AY: 2026-2027

SL No	Week	Topic	Detailed Subtopics
1	Week 2	SVM	SVM Introduction, Applications, Working Principle, Hyperplane, Support Vectors and Margin, Hard and Soft Margin, C Parameter, Kernel Trick, Types of Kernels, Kernel Selection, Advantages and Limitations

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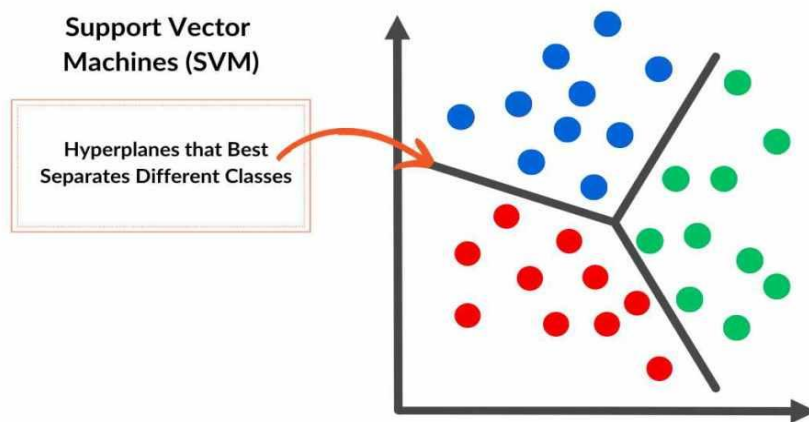
1. INTRODUCTION TO SUPPORT VECTOR MACHINE

1.1 What is SVM?

Support Vector Machine (SVM) is a supervised machine learning algorithm used mainly for classification and also for regression tasks.

The main idea of SVM is to find the best decision boundary that separates different classes of data.

For classification, SVM does not simply find any boundary that separates the classes. It tries to find the boundary with the maximum possible margin.



Example

Suppose we have three classes:

- ● Class A
- ● Class B
- ● Class C

Several lines may separate the three classes.

SVM chooses the line that provides the largest distance from the closest data points of all the classes.

These closest points are called support vectors.

Main objective: Find the decision boundary that separates the classes while maximizing the margin.

1.2 Why Do We Need SVM?

Consider a dataset containing two classes.

There may be many possible decision boundaries:

- Boundary 1 separates the classes but passes very close to the data.
- Boundary 2 separates the classes with a reasonable gap.
- Boundary 3 separates the classes with the largest gap.

SVM chooses the boundary with the **maximum margin**.

Why?

A larger margin generally provides better separation between classes and can help the model generalize better to unseen data.

1.3 Classification and Regression

SVM can be used for:

Classification

The goal is to predict a **class label**.

Examples:

- Spam / Not Spam
- Tumor / No Tumor
- Cat / Dog
- Fraud / Not Fraud

Regression

The goal is to predict a **continuous numerical value**.

Examples:

- House price
- Temperature
- Demand

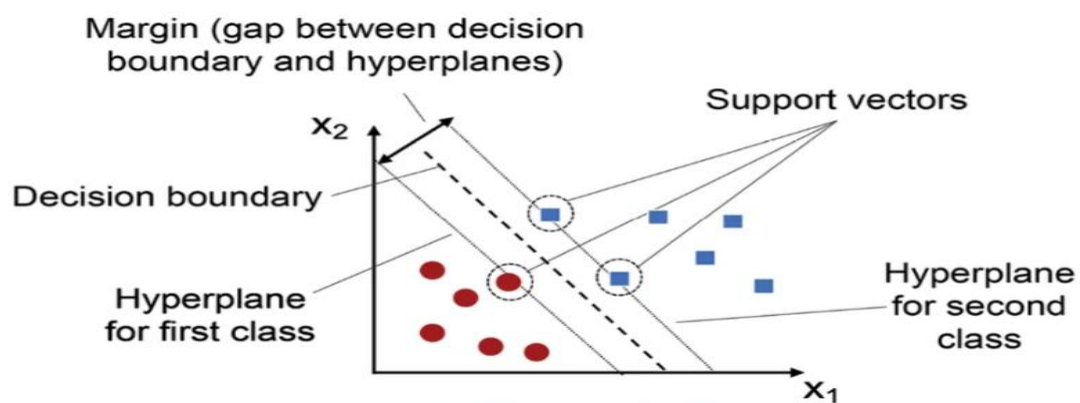
The classification version is generally called **SVC**, while the regression version is called **SVR**.

1.4 Applications of SVM

SVM can be applied to:

- Text classification
- Spam detection
- Sentiment analysis
- Image classification
- Handwriting recognition

2. BASIC CONCEPTS OF SVM



2.1 Feature Space

The **feature space** is the space in which the input data points are represented using their features.

For example, consider two features:

- = Age
- = Salary

Each person can be represented as:

Therefore, the data can be plotted in a two-dimensional feature space.

2.2 Hyperplane

A **hyperplane** is a decision boundary used to separate different classes.

For two-dimensional data, the hyperplane is a **line**.

For three-dimensional data, it is a **plane**.

For higher dimensions, we generally call it a **hyperplane**.

The equation of a linear hyperplane is:

$$w^T x + b = 0$$

where:

- w = weight vector
- x = input feature vector
- b = bias/intercept

Simple idea

The hyperplane divides the feature space into different regions corresponding to different classes.

2.3 Decision Boundary

The hyperplane acts as the **decision boundary**.

For binary classification:

$$w^T x + b > 0$$

may represent one class, while

represents the other class.

$$w^T x + b < 0$$

The exact assignment of positive and negative classes depends on how the labels are defined.

2.4 Support Vectors

Support vectors are the training data points closest to the decision boundary.

They are extremely important because they determine the position and orientation of the optimal hyperplane.

Important point

If some non-support-vector points are removed, the decision boundary may remain almost unchanged.

But if important support vectors are removed, the decision boundary can change significantly.

Remember

Support vectors are the critical data points that "support" the optimal decision boundary.

2.5 Margin

The **margin** is the distance between the decision boundary and the closest data points from each class.

SVM tries to maximize this margin.

For the optimal hyperplane:

$$w^T x + b = 0$$

the two margin boundaries are:

$$w^T x + b = +1$$

and The support vectors lie on these margin boundaries.

$$w^T x + b = -1$$

2.6 Maximum-Margin Principle

Suppose three different hyperplanes can separate the classes:

- L1
- L2
- L3

SVM chooses the hyperplane with the **largest margin**.

Core principle

The goal is not merely: "Separate the classes."

The goal is: "**Separate the classes with the largest possible margin.**"

3. MATHEMATICAL FOUNDATION OF SVM

3.1 Equation of a Hyperplane

The decision boundary is:

$$w^T x + b = 0$$

For two features:

$$w_1 x_1 + w_2 x_2 + b = 0$$

This represents a straight line.

3.2 Classification Function

The SVM decision function is:

$$f(x) = w^T x + b$$

Prediction is based on the sign:

$$\hat{y} = \begin{cases} +1 & f(x) > 0 \\ -1 & f(x) < 0 \end{cases}$$

Thus:

$$\hat{y} = \text{sign}(w^T x + b)$$

3.3 Distance of a Point from the Hyperplane

The perpendicular distance between a point and the hyperplane is:

$$d = \frac{|w^T x + b|}{||w||}$$

For support vectors, the corresponding margin boundaries are:

$$w^T x + b = +1$$

And

$$w^T x + b = -1$$

Therefore, the distance from the decision boundary to either margin is:

$$\frac{1}{||w||}$$

3.4 Margin Calculation

The complete margin is therefore:

$$\text{Margin} = \frac{2}{||w||}$$

This is extremely important.

If $||w||$ is small:

$$\frac{2}{||w||}$$

becomes large.

Therefore:

$$\text{Maximize Margin} \iff \text{Minimize } ||w||$$

3.5 SVM Optimization Objective

For linearly separable data, SVM solves:

$$\min_{w,b} \frac{1}{2} ||w||^2$$

subject to:

$$y_i(w^T x_i + b) \geq 1$$

where:

- x_i = training point
- y_i = class label
- w = weight vector
- b = bias

Why this constraint?

It ensures that the training points are correctly classified and lie on the correct side of the margin.

Simple interpretation

SVM mathematically converts "**find the widest possible margin**" into an optimization problem.

4. HARD-MARGIN SVM

4.1 Linearly Separable Data

If two classes can be perfectly separated by a straight line or hyperplane, the data is called **linearly separable**.

Example:



A boundary can separate all points without errors.

4.2 Hard-Margin Concept

Hard-margin SVM assumes:

- No outliers
- No noisy observations
- Perfectly separable classes
- No training point is allowed to violate the margin

Therefore, the constraints are strict:

$$y_i(w^T x_i + b) \geq 1$$

Main objective

$$\min \frac{1}{2} ||w||^2$$

4.3 Problem with Hard Margin

Real-world data is rarely perfectly clean.

There may be:

- Outliers
- Noise
- Overlapping classes
- Incorrect labels

If one unusual point makes perfect separation impossible, hard-margin SVM may fail to find a valid boundary.

Therefore, we need **soft-margin SVM**.

5. SOFT-MARGIN SVM

5.1 Why Soft Margin?

Soft-margin SVM allows some data points to:

- Enter the margin

- Violate the margin
- In some cases, be misclassified

This makes SVM more suitable for real-world datasets.

5.2 Outliers and Overlapping Classes

Suppose one point from Class A lies inside the region of Class B.

Forcing SVM to classify that one point correctly may produce a very narrow and unstable boundary.

Soft-margin SVM says:

"I can tolerate some violations if doing so gives me a better overall boundary."

5.3 Slack Variables

Soft-margin SVM introduces a variable called the **slack variable**, represented by:

$$\xi_i$$

The constraint becomes:

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

where:

$$\xi_i \geq 0$$

ξ_i	Meaning
0	Correctly classified and outside/on margin
$0 < \xi_i < 1$	Correctly classified but inside margin
$\xi_i = 1$	On decision boundary
$\xi_i > 1$	Misclassified

5.4 Soft-Margin Mathematical Formulation

The objective becomes:

$$\min \left(\frac{1}{2} \|w\|^2 + C \sum_i \xi_i \right)$$

There are two competing objectives:

First term

$$\frac{1}{2} ||w||^2 \rightarrow \text{encourages a large margin}$$

Second term

$$C \sum_i \xi_i$$

→ penalizes **margin violations/errors**

Therefore:

SVM balances margin maximization and error minimization

5.5 Hinge Loss

The hinge loss for an individual training point is:

$$L_i = \max(0, 1 - y_i f(x_i))$$

where:

$$f(x) = w^T x + b$$

Interpretation

If:

$$y_i f(x_i) \geq 1$$

then:

$$L_i = 0$$

The point is correctly classified and outside the margin

If:

$$y_i f(x_i) < 1$$

then the loss becomes positive.

Therefore:

- Correct + outside margin → **zero loss**
- Correct + inside margin → **positive loss**
- Incorrectly classified → **larger loss**

6. C PARAMETER

6.1 What is C?

C is a regularization parameter that controls how strongly SVM penalizes classification errors and margin violations.

The soft-margin objective is:

$$\frac{1}{2} ||w||^2 + C \sum_i \xi_i$$

Therefore, determines how important the error penalty is.

6.2 Small C

When:

SVM allows more margin violations.

The model tends to prefer:

- Wider margin
- Simpler boundary
- More tolerance to errors

Risk: Underfitting

6.3 Large C

When:

SVM strongly penalizes errors.

The model tries harder to classify training points correctly.

It may produce:

- Narrower margin
- More complex boundary
- Greater sensitivity to outliers

Risk: Overfitting

6.4 C and Margin

Small C

"I am willing to tolerate some errors to maintain a wider margin."

Large C

"I strongly want to avoid training errors, even if the margin becomes narrower."

Remember

$C \rightarrow$ Penalty for training errors

7. LINEAR AND NONLINEAR SVM

Sometimes the two groups can be separated with a simple straight line. Other times the groups are mixed in a way that no straight line can separate them this is where kernels come in.

7.1 Linear SVM

If the classes can be separated using a straight line/hyperplane, a **linear kernel** can be used.

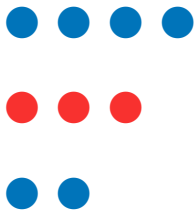
Example:



7.2 Nonlinear SVM

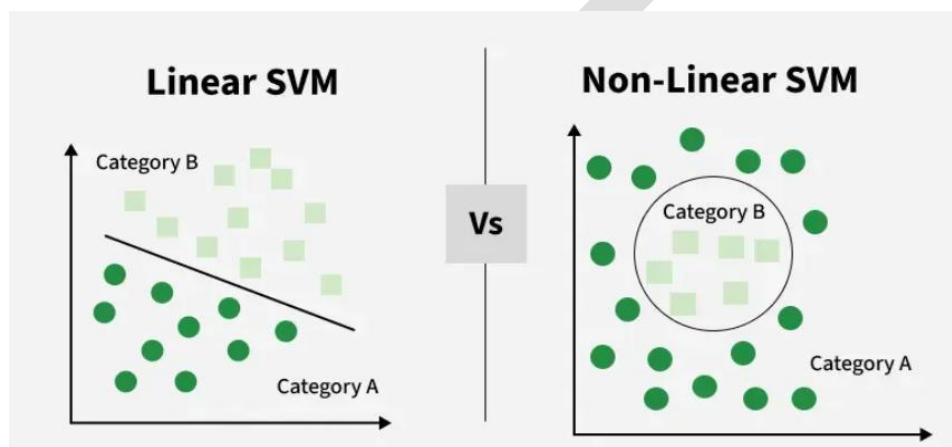
Sometimes the classes cannot be separated using a straight line.

Example:



A curved boundary may be required.

This is where **kernel functions** become useful.



7.3 Why Linear Boundary May Be Insufficient

Consider two classes where:

- One class forms a circle
- Another class surrounds it

No straight line can separate the two classes.

A nonlinear boundary is required.

The **kernel trick** allows SVM to handle such situations.

8. KERNEL TRICK

8.1 Why Do We Need Kernels?

Some datasets are not linearly separable in their original feature space.

A possible solution is to transform the data into a higher-dimensional space where separation becomes easier.

$$x \rightarrow \phi(x)$$

8.2 Mapping to Higher Dimensions

Suppose data exists in 2D and cannot be separated by a straight line.

We can transform it into 3D.

In the higher-dimensional space, a plane may be able to separate the classes.

Then that separation can correspond to a nonlinear boundary in the original space.

8.3 Kernel Trick Intuition

Instead of explicitly calculating the transformation:

$$\phi(x)$$

the kernel calculates the required similarity directly.

This is called the **Kernel Trick**.

Simple explanation

The kernel trick gives SVM the benefits of working in a higher-dimensional feature space without explicitly constructing all the transformed features.

8.4 Mathematical Representation

The kernel function is:

$$K(x, z) = \phi(x)^T \phi(z)$$

Here:

- x, z = input data points
- $\phi(x), \phi(z)$ = transformed representations
- $K(x, z)$ = kernel similarity

8.5 Advantages of Kernel Trick

It allows SVM to:

- Handle nonlinear relationships

- Avoid explicitly creating very large feature spaces
- Work with complex decision boundaries
- Efficiently calculate relationships between data points

9. TYPES OF KERNELS

A kernel is simply a function that decides the shape of the boundary SVM is allowed to draw. Think of each kernel as a different “tool” for drawing that boundary — some tools only draw straight lines, others can draw curves or closed loops.

9.1 LINEAR KERNEL

The linear kernel is:

$$K(x, z) = x^T z$$

It essentially uses the original feature space without introducing nonlinear transformations.

Decision boundary

Straight line / hyperplane.

Use when

- Data is approximately linearly separable
- Number of features is very large
- Text classification
- Feature space is already sufficiently expressive

Advantages

- Simple
- Fast

Limitation

Cannot model highly nonlinear relationships.

9.2 POLYNOMIAL KERNEL

The polynomial kernel is commonly written as:

$$K(x, z) = (\gamma x^T z + r)^d$$

where:

- = degree
- = scaling parameter
- = independent constant, commonly represented by coef0

What does it do?

It allows SVM to model:

- Feature interactions
- Curved boundaries
- Polynomial relationships

For example, with:

- Glucose
- BMI

the model can capture relationships involving:

Advantages

- Handles nonlinear data
- Captures feature interactions

Disadvantages

- Computationally more expensive
- High degree may cause overfitting

9.3 RBF / GAUSSIAN KERNEL

The RBF kernel is:

$$K(x, z) = e^{-\gamma \|x - z\|^2}$$

RBF measures similarity based on the distance between data points.

Basic idea

If two points are close:

$$\|x - z\| \rightarrow \text{small}$$

then their similarity is high.

$$\|x - z\| \rightarrow \text{large}$$

If they are far apart:

then their similarity is low.

Gamma

Gamma controls how quickly the influence decreases.

High Gamma

- Local influence
- Complex boundary
- Overfitting risk

Low Gamma

- Wider influence
- Smooth boundary
- Underfitting risk

9.4 SIGMOID KERNEL

The sigmoid kernel is commonly written as:

$$K(x, z) = \tanh(\gamma x^T z + r)$$

It is inspired by the sigmoid-like activation functions used in neural networks.

Parameters

- C
- γ
- coef0

Advantages

- Can model nonlinear relationships
- Conceptually related to neural-network activation

Limitations

- Less commonly used
- May not perform as well as RBF for many datasets

9.5 IMPORTANT KERNEL PARAMETERS

Gamma

Gamma is particularly important for nonlinear kernels such as RBF.

It controls:

How strongly an individual training point influences the decision boundary.

9.6 Effect of Gamma

Low Gamma

Each point has a wider area of influence.

Result:

- Smooth boundary
- Simpler model
- Possible underfitting

High Gamma

Each point has a very local influence.

Result:

- Complex boundary
- Closely follows training data
- Possible overfitting
- **High gamma:** influence is very short-range → boundary wraps tightly around individual points → risk of overfitting.
- **Low gamma:** influence reaches far → boundary becomes smooth and simple → risk of underfitting.

9.7 Gamma and Sigma

RBF can also be written as:

$$K(x, z) = e^{-\frac{\|x-z\|^2}{2\sigma^2}}$$

Therefore:

$$\gamma = \frac{1}{2\sigma^2}$$

Thus:

Large gamma ↔ Small sigma

More local influence.

Small gamma ↔ Large sigma

Wider influence.

9.8 coef0

coef0 is the independent constant used in Polynomial and Sigmoid kernels.

It affects the contribution of the constant term in the kernel calculation.

It is generally less important for the RBF kernel because RBF does not use this parameter.

9.9 C AND GAMMA TOGETHER

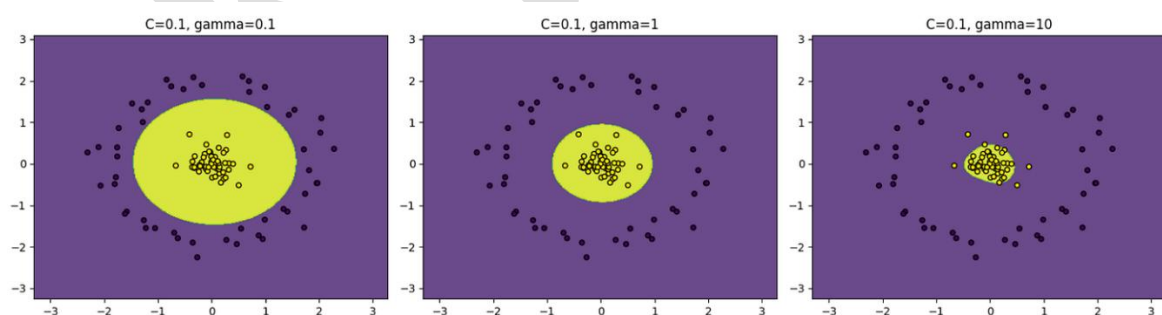
This is an important concept.

C controls:

How strongly training errors are penalized

Gamma controls:

How local the influence of each point is

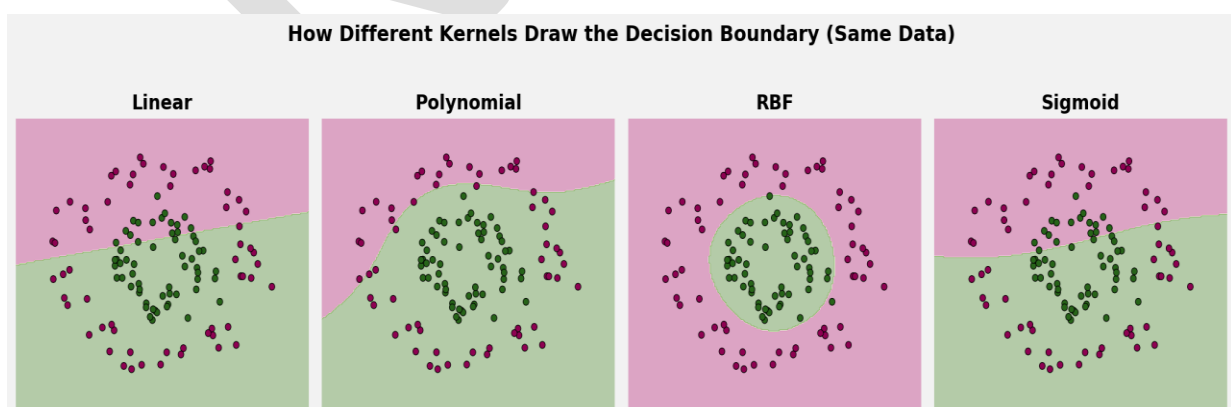


Parameter	What does it control?	Small Value	Large Value	Key Idea
C	Controls how strongly SVM	Allows more errors, gives a wider margin,	Penalizes errors strongly, tries to classify training points correctly,	C controls how much SVM cares

Parameter	What does it control?	Small Value	Large Value	Key Idea
	penalizes training errors	simpler model, may cause underfitting	more complex model, may cause overfitting	about training errors.
Gamma (γ)	Controls how strongly an individual data point influences the decision boundary	Wider influence, smoother and simpler boundary, may cause underfitting	More local influence, complex boundary, may cause overfitting	Gamma controls how sensitive the model is to input data.
coef0	Controls the independent constant term in Polynomial and Sigmoid kernels	Lower contribution of the constant term	Higher contribution of the constant term; can change the shape/position of the kernel transformation	coef0 controls the effect of the constant term in the kernel.

9.10 Seeing All Four Kernels Side by Side

The picture below shows the same dataset one class forming a ring around another class classified using each of the four kernels. Notice how only RBF (and to some extent Polynomial) can properly wrap around the inner class; Linear and Sigmoid are forced into a straight-ish cut that misses many points.



9.11 Choosing the Right Kernel

There is no single kernel that works best for every dataset. A simple decision guide:

Situation	Recommended Kernel	Reason
Data is roughly linearly separable, or features $\geq$ samples (e.g., text/NLP)	Linear	Simple, fast, avoids overfitting when features already carry enough information.
Moderate curvature, feature interactions matter	Polynomial	Captures interaction terms between features explicitly.
Boundary shape is unknown / classes form clusters, rings, or irregular regions	RBF (Gaussian)	Most flexible; strong default choice in most cases.
Neural-network-like behaviour desired (rare)	Sigmoid	Mimics a shallow neural network activation; rarely the best performer.

- In practice, RBF is the most commonly used default kernel, and its gamma and C parameters are usually tuned together using grid search with cross-validation.

Practical rule

If you don't know the shape of the boundary:

Then parameters such as **C** and **gamma** can be tuned.

10. FEATURE SCALING IN SVM

10.1 Why is Scaling Important?

SVM, especially with RBF and Polynomial kernels, can be sensitive to the scale of input features.

Consider:

- Age: 20–60
- Salary: 20,000–2,00,000

Salary has much larger numerical values.

If distance-based calculations are used, salary can dominate the calculation.

10.2 Effect on RBF

RBF uses:

Therefore, feature magnitudes directly influence the distance.

If one feature has a much larger scale, it can dominate.

Therefore:

Features should generally be brought to comparable scales before applying SVM.

11. SUPPORT VECTOR REGRESSION

11.1 SVM for Regression

SVM is not limited to classification.

For regression, we use:

Instead of finding a boundary between classes, SVR tries to find a function that predicts continuous values.

11.2 SVC vs SVR

SVC	SVR
Classification	Regression
Predicts classes	Predicts numerical values
Decision boundary	Regression function
Example: spam/not spam	Example: house price

12. ADVANTAGES AND LIMITATIONS OF SVM

12.1 Advantages

1. Effective in high-dimensional spaces

SVM works well when the number of features is large.

2. Good for nonlinear data

Kernel functions allow nonlinear decision boundaries.

3. Maximum-margin principle

The model focuses on finding a boundary with a strong separation margin.

4. Works for classification and regression

SVC can be used for classification and SVR for regression.

5. Useful for relatively small/medium-sized datasets

SVM can be effective when the dataset is not extremely large.

6. Flexible

Different kernels allow different types of decision boundaries.

12.2 Limitations

1. Training can be expensive

SVM may become computationally expensive for very large datasets.

2. Parameter tuning

Selecting:

- Kernel
- C
- Gamma
- Degree

can require careful tuning.

3. Feature scaling is important

Especially for RBF and Polynomial kernels.

4. Sensitive to noisy/overlapping data

A poor choice of C and kernel can produce poor generalization.

5. Interpretability

Complex nonlinear kernels can be difficult to interpret.

13. APPLICATIONS OF SVM

Text Classification

- Spam detection
- Sentiment analysis
- Document classification

Image Classification

- Image recognition
- Handwriting recognition
- Object/category classification

Medical Applications

- Tumor classification
- Disease classification
- Medical image analysis

Anomaly Detection

- Fraud detection
- Intrusion detection
- Fault detection

14. COMPLETE SVM CONCEPTUAL FLOW

Step 1 — Start with the data

Input features are represented in feature space.

↓

Step 2 — Find a decision boundary

SVM searches for a hyperplane:

↓

Step 3 — Identify support vectors

Find the points closest to the boundary.



Step 4 — Maximize the margin



Step 5 — Handle imperfect data

If the data contains overlap/outliers, use **soft margin**.



Step 6 — Introduce slack variables

allow margin violations.



Step 7 — Control errors using C



Step 8 — If data is nonlinear

Use a **kernel**.



Step 9 — Select a kernel

- Linear
- Polynomial
- RBF
- Sigmoid



Step 10 — Tune kernel parameters

Especially:



Step 11 — For regression/Classification

Use: SVR/SVC

Interpretation & Analysis Questions

Q1. Choosing Between Two Decision Boundaries

A dataset contains two classes that can be separated by several straight lines. One line passes very close to the training points, while another leaves a large gap between the two classes. Both classify all training samples correctly.

Which boundary should SVM prefer? Explain why.

Answer:

SVM should prefer the boundary with the **larger margin**. SVM does not simply look for a boundary that correctly classifies the training data; it tries to maximize the distance between the decision boundary and the nearest data points. A larger margin generally provides better separation and can improve generalization.

Q2. Effect of an Outlier

A dataset contains two clearly separated classes. One point belonging to Class A is located inside the region occupied by Class B. If a model is forced to classify this point correctly, the decision boundary becomes very narrow and irregular.

What SVM approach would be more appropriate and why?

Answer:

A **soft-margin SVM** would be more appropriate. It allows some margin violations or even misclassified points instead of forcing the boundary to fit every training point. This can produce a wider and more robust boundary in the presence of outliers or noisy data.

Q3. Choosing the Value of C

Two SVM models are trained on the same noisy dataset.

- Model A uses
- Model B uses

Model B classifies almost every training point correctly, but its boundary is very sensitive to individual noisy observations.

Which model is more likely to overfit? Explain.

Answer:

Model B with is more likely to overfit. A large C strongly penalizes training errors, so the SVM tries very hard to classify training points correctly. This can result in a narrower and more complex boundary that becomes sensitive to noise and outliers.

Q4. Selecting a Kernel

You are given a dataset where one class forms a small circular cluster in the centre and another class surrounds it like a ring. A straight line cannot separate the two classes.

Would you choose a Linear kernel or RBF kernel? Why?

Answer:

The **RBF kernel** would be a better choice. The classes have a nonlinear, circular structure, so a straight-line boundary from a Linear kernel cannot separate them effectively. RBF can create flexible nonlinear boundaries and can form a closed boundary around the inner class.

Q5. Interpreting Gamma

An RBF SVM initially produces a smooth decision boundary. After increasing gamma substantially, the boundary begins to curve tightly around individual training points.

What does this indicate about the effect of gamma? What problem might occur?

Answer:

A high gamma means that each training point has a **very local influence**. The decision boundary becomes more complex and closely follows individual observations. This increases the risk of **overfitting**, where training performance is high but performance on unseen data may decrease.

Q6. C vs Gamma

Two students make the following statements:

- Student A: “ C controls how much SVM penalizes classification errors.”
- Student B: “Gamma controls how strongly individual points influence the decision boundary.”

Are both students correct? Explain the difference between C and gamma.

Answer:

Yes, both are correct.

- **C** controls the penalty assigned to training errors and margin violations.
- **Gamma** controls the range of influence of individual data points, particularly in the RBF kernel.

Q7. Why Scaling Changes the Result

A dataset contains two features:

- Age: values between 20 and 60
- Annual income: values between 20,000 and 2,00,000

An RBF-based SVM performs poorly before scaling but improves significantly after the features are scaled.

Why can feature scaling make such a difference?

Answer:

The RBF kernel uses the distance between data points:

Since income has a much larger numerical range than age, it can dominate the distance calculation. Scaling puts the features on comparable ranges, allowing the kernel to consider the features more appropriately.

Q8. Identifying the Important Data Points

After training an SVM, a student notices that only a small subset of the training observations lies very close to the decision boundary.

Why are these observations particularly important to the SVM? What are they called?

Answer:

These observations are called **support vectors**. They are important because they determine the position and orientation of the optimal decision boundary. Points far away from the boundary generally have less direct influence on its location.